SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: B.E / B.Tech

Course Code: MLA02

Course Name: FUNDAMENTALS OF MACHINE LEARNING

Duration: 1:30 Hr

Date : 31-10-2026

Max Marks: 50 Marks

ASSIGNMENT – I

UNIT I – INTRODUCTION AND MATHEMATICAL FOUNDATIONS

Topics: *Design of a learning system, types of machine learning, probability theory and distributions, Bayes decision theory, information theory*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO1	Analyse machine learning fundamentals and mathematical foundations, including probability theory, Bayes decision theory, and information theory.

Answer all the Tasks

Task 1. A retail bank plans five machine-learning systems. For each system, analyse the problem and state (i) the learning paradigm (supervised / unsupervised / semi-supervised / reinforcement), (ii) the learning task (classification, regression, clustering, control, etc.) and (iii) the experience E and performance measure P in Mitchell's (T, P, E) formulation:

(a) Predict whether a loan applicant will default, using ten years of labelled repayment records.

(b) Discover behavioural customer segments among 2 million customers for which no labels exist.

(c) Estimate the next-month credit-card spend (in ₹) of each customer.

(d) Learn an ATM cash-refill policy that maximises cash availability minus transport cost through repeated trial and feedback.

(e) Only 2% of 500,000 transaction records are labelled fraud/genuine; the rest are unlabelled. Build a fraud detector that exploits both kinds of data.

Finally, justify which of the five systems is most harmed by label noise.

(5) (CO1) (BL4)

Task 2. A screening test for a disease with prevalence 2% has sensitivity 95% and specificity 90%. Calculate the probability that a person who tests positive actually has the disease, and analyse why the answer is much lower than the sensitivity.

(2) (CO1) (BL4)

Task 3. The joint probability mass function of a discrete random vector (X, Y) is given below.

X \ Y	Y = 0	Y = 1
X = 0	0.10	0.20
X = 1	0.15	0.25

X \ Y	Y = 0	Y = 1
X = 2	0.20	0.10

Analyse the distribution by (a) finding the marginals of X and Y, (b) computing E[X], Var(X) and E[Y], (c) computing $P(X = 2 \mid Y = 1)$, (d) computing Cov(X, Y) and the correlation coefficient, and (e) deciding whether X and Y are independent, with a justification.

(6) (CO1) (BL4)

Task 4. A temperature sensor output is modelled as $X \sim N(\mu = 50, \sigma^2 = 16)$.

(a) Calculate $P(X > 58)$ and $P(46 < X < 54)$.

(b) A second sensor gives the readings 12, 15, 11, 14, 13. Derive the maximum-likelihood estimates of its mean and variance and compare the ML variance with the unbiased estimate, stating which is biased and why. (Use $\Phi(1) = 0.8413$, $\Phi(2) = 0.9772$.)

(4) (CO1) (BL4)

Task 5. A source emits four symbols with probabilities (1/2, 1/4, 1/8, 1/8). Calculate its entropy, compare it with the entropy of a uniform source over four symbols, and analyse the redundancy. Also compute the average length of the code {0, 10, 110, 111} and comment on its optimality.

(3) (CO1) (BL4)

Task 6. A model predicts class probabilities Q = (1/3, 1/3, 1/3) while the true distribution is P = (0.5, 0.25, 0.25). Compute KL(P||Q) and KL(Q||P) in bits, and analyse what the asymmetry implies when KL divergence is used as a training loss.

(3) (CO1) (BL4)

Task 7. Two binary variables A and B have the joint distribution $P(0,0) = 0.4$, $P(0,1) = 0.1$, $P(1,0) = 0.1$, $P(1,1) = 0.4$. Calculate H(A), H(B), the joint entropy H(A,B), the conditional entropy H(A|B) and the mutual information I(A;B). Analyse what the value of I(A;B) says about using B as a feature to predict A.

(4) (CO1) (BL4)

Task 8. A one-dimensional feature x follows $p(x|\omega_1) = N(2, 4)$ for class ω_1 and $p(x|\omega_2) = N(6, 4)$ for class ω_2, with priors $P(\omega_1) = 0.6$ and $P(\omega_2) = 0.4$ and zero–one loss. Derive the Bayes decision threshold x^*, classify $x = 4.5$, and calculate the overall probability of error. Analyse how the threshold moves as the prior of ω_1 increases.

(4) (CO1) (BL4)

Task 9. For an observation x, the posteriors are $P(\omega_1|x) = 0.3$ (disease) and $P(\omega_2|x) = 0.7$ (healthy). Action α_1 = 'declare disease', α_2 = 'declare healthy'. The loss matrix is $\lambda(\alpha_1|\omega_1) = 0$, $\lambda(\alpha_1|\omega_2) = 8$ (unnecessary treatment), $\lambda(\alpha_2|\omega_1) = 50$ (missed disease), $\lambda(\alpha_2|\omega_2) = 0$. Evaluate the conditional risks, decide the minimum-risk action, compare it with the MAP decision and determine the posterior $P(\omega_1|x)$ above which α_1 is optimal.

(3) (CO1) (BL5)

Task 10. Polynomial regression models of increasing degree were fitted to the same dataset, giving the mean-squared errors below.

Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1

Degree	1	2	3	5	9
Validation MSE	9.5	4.6	2.9	3.6	11.8

Calculate the generalisation gap for each degree, diagnose under-fitting/over-fitting using the bias–variance trade-off, select the best model with justification, and recommend two techniques to improve the degree-9 model.

(4) (CO1) (BL5)

Task 11. A coin is tossed 10 times and shows 7 heads. The bias θ = P(head) is given a Beta(2, 2) prior. Calculate the ML estimate, the MAP estimate and the posterior mean of θ. Then evaluate how the three estimates change if the data were 70 heads in 100 tosses, and explain the role of the prior.

(4) (CO1) (BL5)

Task 12. A hospital wants an early-warning system that raises an alert when an ICU patient is likely to develop sepsis within the next 6 hours. Historical electronic health records (vital signs, lab values, outcomes) are available.

(a) Design the learning system: state the task T, performance measure P and experience E, the target function, a suitable input representation, a learning algorithm and the validation strategy. [3]

(b) On a validation cohort of 1,000 patients, 5% develop sepsis. The trained model has sensitivity 0.90 and specificity 0.80. Compute TP, FN, FP and TN. [1]

(c) Compute the positive predictive value (PPV) and negative predictive value (NPV). [1]

(d) A missed sepsis case costs ₹100 (unit cost) and a false alarm costs ₹5. Compare the total cost of (i) using the model, (ii) alerting every patient and (iii) never alerting. [2]

(e) Recommend a deployment policy and justify how the alert threshold should be set using Bayes decision theory. [1]

(8) (CO1) (BL6)

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ANSWER SCRIPT EVALUATION SHEET

ASSIGNMENT - I

Programme	B.E / B.Tech	Course Code	MLA02
Course Name	Fundamentals of Machine learning.	Date	31-10-2026
Duration	1 Hr 30 Min	Max Marks	50

Register No.: P. Surendra Kumar Reddy (192425224)

TASK 1	Max Marks	Marks Awarded
Task 1	5	5
Task 2	2	2
Task 3	6	6
Task 4	4	4
Task 5	3	3
Task 6	3	3
Task 7	4	4
Task 8	4	4
Task 9	3	3
Task 10	4	4
Task 11	4	4
Task 12	8	8
TOTAL	**50**	50

Total in words: fifty Marks

50/50

ASSIGNMENT - I

Name : P Surendra Kumar Reddy

Reg NO : 19242S224

Course Name : Fundamentals of Machine Learning.

Course Code : MLA0205

Task 1. Analysis of 5 Machine Learning Systems

(a). Loan Default prediction

- Learning paradigm: Supervised Learning
- Learning Task(T) Classification (predicting whether a customer will default or not).
- Experience (E): Ten years of labelled loan repayment records.
- performance measure (P): Accuracy, precision, recall and F_1 - score.

(b) Customer Behaviour segmentation

- Learning paradigm: Unsupervised Learning
- Learning Task (T): clustering (grouping customers based on similar behaviour).
- Experience (E): Data of 2 millions customers without labels
- performance (P): silhouette score and cluster quality.

(c). Credit card spending prediction

- Learning paradigm: Supervised Learning
- Learning Task (T): Regression (predicting next-month spending in rupees).
- Experience(E): Historical customer spending records with actual spending values.
- performance (P): Mean Absolute Error (MAE) and Mean squarred Error (MSE).

(d) ATM Cash-Refill policy.

- Learning paradigm: Reinforment Learning
- Learning Task (T): Sequential decision-making & Control.
- Experience (E): Repeated trails, actions and feedback

from ATM operations.
• performance (P): maximum cash availability and min transportation cost.

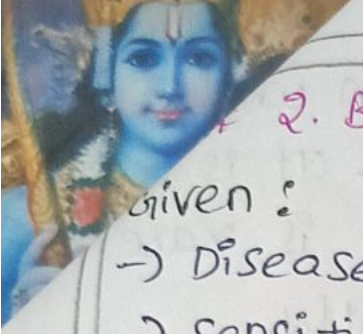
(e) Fraud detection with limited labels.

• Learning paradigm: semi-supervised learning
• Learning Task (T): classification
• Experience (E): 2% labelled and 98% unlabelled transaction records.
• performance (P): accuracy, precision, recall & F1-score.

Justification: which system is most harmed by label noise?

* The loan default prediction system is highly affected by label noise because it depends on labelled repayment records
* Incorrect labels can cause the model to learns wrong patterns and make incorrect predictions.
* The fraud detection system is also sensitive because only 2% of its records are labelled.

Conclusion: supervised learning systems are particularly sensitive to label noise because they directly learn from labelled examples.



2. Baye's Theorem - Disease screening

Given :

→ Disease prevalence, $p(D) = 0.02$

→ Sensitivity, $p(+|D) = 0.95$

→ Specificity $= 0.90$

→ false positive rate, $p(+|\bar{D}) = 1 - 0.90 = 0.10$

Formula :

Using Baye's theorem,

$$p(D|+) = \frac{p(+|D)\ p(D)}{p(+)}$$

$$p(D|+) = \frac{p(+|D)\ p(D)}{p(+|D)\,p(D) + p(+|\bar{D})\,p(\bar{D})}$$

Substituting the values,

$$p(D|+) = \frac{0.95 \times 0.02}{(0.95 \times 0.02) + (0.10 * 0.98)}$$

$$= \frac{0.019}{0.019 + 0.098} = \frac{0.019}{0.117} = 0.1624$$

$p(D|+) = 16.24\ \%$

Analysis :

* The sensitivity is 95%, which means the correctly identifies 95% of diseased people.
* However, disease prevalence is only 2%.
* Therefore, false positives from the much larger healthy population significantly affect the result.

Final Answer: the probability that a person ... positive actually has the disease is 16.24%. It ... lower than sensitivity because the disease is rare and false positives contribute significantly.

Task 3. Joint probability distribution

Given Joint probability mass function

X|Y	Y=0	Y=1
X=0	0.10	0.20
X=1	0.15	0.25
X=2	0.20	0.10

(a) Find the marginal probabilities of X and Y.

marginal probability of X:

$$P(X=x) = \sum_y P(X=x, Y=y)$$

$P(X=0) = 0.10 + 0.20 = \boxed{0.30}$

$P(X=1) = 0.15 + 0.25 = \boxed{0.40}$

$P(X=2) = 0.20 + 0.10 = \boxed{0.30}$

marginal probability of Y:

$$P(Y=y) = \sum_x P(X=x, Y=y)$$

$P(Y=0) = 0.10 + 0.15 + 0.20 = \boxed{0.45}$

$P(Y=1) = 0.20 + 0.25 + 0.10 = \boxed{0.55}$

Marginal distributions:

X	P(X)	Y	P(Y)
0	0.30	0	0.45
1	0.40	1	0.55
2	0.30		



Calculate $E[X]$, $Var[X]$ and $E[Y]$.

expected value of X:

$$E[X] = \sum x\, p(X=x)$$
$$= (0 \times 0.30) + (1 \times 0.40) + 2(0.30)$$
$$= 0 + 0.40 + 0.60$$

$$\boxed{E[X] = 1}$$

variance of X:

$$Var(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = (0^2 \times 0.30) + (1^2 \times 0.40) + (2^2 \times 0.30)$$
$$= 0 + 0.40 + 1.20 = 1.60$$

$$Var(X) = 1.60 - (1)^2 = 0.60$$

$$\boxed{Var(X) = 0.60}$$

expected value of Y:

$$E[Y] = \sum y\, P(Y=y)$$
$$= (0 \times 0.45) + (1 \times 0.55)$$

$$\boxed{E[Y] = 0.55}$$

(c) calculate $P(X=2 \mid Y=1)$.

formula:

$$P(X=2 \mid Y=1) = \frac{P(X=2, Y=1)}{P(Y=1)} = \frac{0.10}{0.55}$$

$$\boxed{P(X=2 \mid Y=1) = 0.1818}$$

(d) calculate Covariance and Correlation Coefficient.

Covariance formula:

$$\boxed{Cov(X, Y) = E[X.Y] - E[X]E[Y]}$$

first calculate $E[XY]$

$$\boxed{E[XY] = \sum_x \sum_y xy\, p(x, y)}$$

$= (0\times0\times0.10) + (0\times1\times0.20) + (1\times0\times0.15) + (1\times1\times0.30) + (2\times0\times0.20) + (2\times1\times0.10)$

$= 0 + 0 + 0 + 0.25 + 0 + 0.25$

$\boxed{E[XY] = 0.45}$

Therefore,

$$Cov(X,Y) = 0.45 - (1\times0.55)$$

$$\boxed{Cov(X,Y) = -0.10}$$

Correlation coefficient

$$r = \frac{Cov(X,Y)}{\sqrt{Var(X)\,Var(Y)}}$$

first, $Var(Y) = 0.55(1-0.55) = 0.2475$

$$r = \frac{-0.10}{\sqrt{0.60\times0.2475}} = \frac{-0.10}{0.3853} = -0.2595$$

$$\boxed{r = -0.2595}$$

(e) check whether X and Y are independent

for independence, $P(X,Y) = P(X)\,P(Y)$

Consider :

$$P(X=0, Y=0) = 0.10$$

But, $P(X=0)\,P(Y=0) = 0.30\times0.45 = 0.135$

$$P(X=0)\,P(Y=0) = 0.135$$

since, $0.10 \neq 0.135$

final Answer : X and Y are not independent because their joint probability is not equal to the product of their marginal probabilities for this pair. The correlation is negative, indicating a weak negative linear relationship.

4. Normal Distribution and maximum Likelihood Estimation

Given: $X \sim N(\mu = 50, \sigma^2 = 16)$

therefore, $\mu = 50$, $\sigma = \sqrt{16} = 4$

(a) calculate $p(X > 58)$ and $p(46 < X < 54)$.

Standard normal formula: $\boxed{Z = \frac{X - \mu}{\sigma}}$

1. calculate $p(X > 58)$:

$$Z = \frac{58 - 50}{4} = \frac{8}{4} = 2$$

$$p(X > 58) = p(Z > 2)$$
$$= 1 - \phi(2)$$
$$= 1 - 0.9772$$

$$\boxed{p(X > 58) = 0.0228}$$

2. calculate $p(46 < X < 54)$:

for $X = 46$

$$Z_1 = \frac{46 - 50}{4} = -1$$

for $X = 54$

$$Z_2 = \frac{54 - 50}{4} = 4$$

$$p(46 < X < 54) = p(-1 < Z < 1)$$
$$= \phi(1) - \phi(-1)$$
$$= 0.8413 - 0.1587$$

$$\boxed{p(46 < X < 54) = 0.6826}$$

(b) Find the MLE of mean and variance

Given sensor readings: 12, 15, 11, 14, 13

Number of observations $n = 5$

step 1: calculate MLE of mean

$$\hat{\mu} = \frac{\sum x_i}{n} = \frac{12+15+11+14+13}{5}$$

$$= \frac{65}{5}$$

$$\boxed{\hat{\mu} = 13}$$

Step 2: calculate MLE of variance

$$\boxed{\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \hat{\mu})^2}$$

x_i	$x_i - 13$	$(x_i - 13)^2$
12	−1	1
15	−2	4
11	−2	4
14	1	1
13	0	0
total		10

$$\hat{\sigma}^2 = \frac{10}{5} = 2$$

$$\boxed{\sigma^2 = 2}$$

step 3: calculate unbiased variance

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{10}{5-1} = \frac{10}{4}$$

$$\boxed{s^2 = 2.5}$$

Comparision:

→ ML variance = 2.

→ unbiased variance = 2.5.

→ ML variance is biased because it divides by n instead of n−1.

→ It tends to understimate the true population variance.

final answer: MLE mean = 13, MLE variance = 2 and unbiased variance = 2.5.

5. Entropy and average code length

Given symbol probabilities:

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$$

Step 1: Calculate entropy

formula: $H(X) = -\sum P(x) \log_2 P(x)$

substituting, $H(X) = -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right]$

$$= -\left[\frac{1}{2}(-1) + \frac{1}{4}(-2) + \frac{1}{8}(-3) + \frac{1}{8}(-3)\right]$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$H(X) = 1.75 \text{ bits}$$

Step 2: Compare with uniform source entropy.

for four equally likely symbols,

$$H(X) = \log_2 4$$

$$H(X) = 2 \text{ bits}$$

Comparision:

- Given source entropy = 1.75 bits
- uniform source entropy = 2 bits

Step 3: calculate redundancy

$$R = H_{max} - H(X)$$

$$= 2 - 1.75$$

$$R = 0.25 \text{ bits}$$

percentage redundancy:

$$= \frac{0.25}{2} \times 100$$

$$\Rightarrow 12.5\%$$

Step 4: calculate average code length

Given code:

symbol probability	code	length
1/2	0	1
1/4	10.	2
1/8	110	3
1/8	111	3

formula:

$$L = \sum p(x_i) l_i$$

$$= \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3)$$

$$= 0.5 + 0.5 + 0.375 + 0.375$$

$$L = 1.75 \text{ bits}$$

Final Analysis:

- Entropy = 1.75 bits
- Average code length = 1.75 bits
- The code is a prefix code because no codeword is the prefix of another.
- It achieves the entropy lower bound and is optimal.

Task 6. KL Divergence

Given:

True distribution: $p = (0.5, 0.25, 0.25)$

predicted distribution: $q = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Step 1: calculate $KL(p||q)$.

formula:

$$D_{KL}(p||q) = \sum_i p_i \log_2 \frac{p_i}{q_i}$$

sub,

$$= 0.5 \log_2 \frac{0.5}{1/3} + 0.25 \log_2 \frac{0.25}{1/3} + 0.25 \log_2 \frac{0.25}{1/3}$$

$= 0.5 \log_2(1.5) + 0.5 \log_2(0.75)$

$= 0.5(0.585) + 0.5(-0.415) = 0.2925 - 0.2075$

$D_{KL}(P||Q) = 0.085$ bits

Step 2: Calculate $KL(Q||P)$.

$$D_{KL}(Q||P) = \sum_i Q_i \log_2 \frac{Q_i}{P_i}$$

$$= \frac{1}{3}\log_2 \frac{1/3}{0.5} + \frac{1}{3}\log_2 \frac{1/3}{0.25} + \frac{1}{3}\log_2 \frac{1/3}{0.25}$$

$$= \frac{1}{3}\left[\log_2(2/3) + 2\log_2(4/3)\right]$$

$$= \frac{1}{3}[-0.585 + 2(0.415)] = \frac{0.245}{3}$$

$D_{KL}(Q||P) = 0.0817$ bits.

Step 3: Analyse the asymmetry

- KL divergence measures the difference between two probability distributions.
- $KL(P||Q)$ and $KL(Q||P)$ have different values.
- Therefore, KL divergence is not symmetric.
- The direction of KL divergence affects the training loss and the way a model is penalised for prediction errors.

Final Answers:

$D_{KL}(P||Q) = 0.085$ bits

$D_{KL}(Q||P) = 0.0817$ bits

KL divergence is assymetric, so reversing the divergence value.

Task 7. Entropy and mutual Information

Given:

$p(0,0) = 0.4$, $p(0,1) = 0.1$, $p(1,0) = 0.1$, $p(1,1) = 0.4$

Step 1: Calculate H(A).

marginal probabilities:

$p(A=0) = 0.4 + 0.1 = 0.5$

$p(A=1) = 0.1 + 0.4 = 0.5$

formula: $H(A) = -\sum p(a) \log_2 p(a)$

$= -[0.5 \log_2 0.5 + 0.5 \log_2 0.5]$

$H(A) = 1$ bit

Step 2: calculate H(B)

similarly, $p(B) = 0.4 + 0.1 = 0.5$

$p(B=1) = 0.1 + 0.4 = 0.5$

$H(B) = -[0.5 \log_2 0.5 + 0.5 \log_2 0.5]$

$H(B) = 1$ bit

Step 3: calculate joint entropy H(A,B).

formula: $H(A,B) = -\sum p(a,b) \log_2 p(a,b)$

$= -[0.4 \log_2 0.4 + 0.1 \log_2 0.1$

$+ 0.1 \log_2 0.1 + 0.4 \log_2 0.4]$

$= -[0.8 \log_2 4 + 0.2 \log_2 0.1]$

$H(A,B) = 1.7219$ bits.

Step 4: calculate Conditional entropy H(A/B)

formula: $H(A/B) = H(A,B) - H(B)$

$= 1.7219 - 1$

$H(A/B) = 0.7219$ bits

s: Calculate mutual information $I(A,B)$.

formula:

$$I(A,B) = H(A) - H(A|B)$$

$$= 1 - 0.7219$$

$$I(A,B) = 0.2781 \text{ bits}$$

Analysis:

- Mutual information measures how much information B provides about A.
- Here, $I(A,B) = 0.2781$ bits.
- Therefore, B provides some useful information for predicting A.
- However, B does not completly determine A.

final Answer: Mutual Information is 0.2781 bits, indicating a weak but useful relationship between A and B.

TASK 8. Bayes Decision Theory

Given:

$$p(x|\omega_1) = N(2,4)$$

$$p(x|\omega_2) = N(6,4)$$

$$p(\omega_1) = 0.6$$

$$p(\omega_2) = 0.4$$

Loss = zero-one loss

Step 1: Desive the Bayes decision threshold.

under zero-one loss, choose the class with their larger posterior probability.

At the dece sion threshold,

$$p(\omega_1)\, p(x|\omega_1) = p(\omega_2)\, p(x|\omega_2)$$

using the guassian density,

$$0.6 \exp\left[-\frac{(x-2)^2}{8}\right] = 0.4 \exp\left[-\frac{(x-6)^2}{8}\right]$$

Taking natural logarithms,

$$\ln(0.6) - \frac{(x-2)^2}{8} = \ln(0.4) - \frac{(x-6)^2}{8}$$

Rearranging, $\ln \frac{0.6}{0.4} = \frac{(x-2)^2 - (x-6)^2}{8}$

$$\ln(1.5) = x - 4$$

Therefore, $x = 4 + \ln(1.5)$

$$\boxed{x^* = 4.4055}$$

Decision rule:

- If x is less than 4.4055, choose class ω_1.
- If x is greater than 4.4055, choose class ω_2.

Step 2: Classify $x = 4.5$

Given, $x = 4.5$

Since, $4.5 > 4.4055$

classify as ω_2

Step 3: calculate overall probability of error.

formula:

$$\boxed{P(\text{error}) = P(\omega_1)\, P(\text{error}/\omega_1) + P(\omega_2)\, P(\text{error}/\omega_2)}$$

for class ω_1:

$$z_1 = \frac{4.4055 - 2}{2} = 1.20275$$

$$\boxed{P(\text{error}/\omega_1) = P(z > 1.20275) \approx 0.1146}$$

for class ω_2:

$$z_2 = \frac{4.4055 - 6}{2} = -0.79725$$

$$\boxed{P(\text{error}/\omega_2) = P(z < -0.79725) \approx 0.2126}$$

Therefore, $P(\text{error}) = 0.6(0.1146) + 0.4(0.2126)$

$$= 0.06876 + 0.08504 = 0.1538$$

$$\boxed{P(\text{error}) = 15.38\ \%}$$

u: Analyse the effect of increasing prior probability.

→ when the prior probability of w_1 increases, the model becomes more likely to select w_1.
→ therefore, the decision threshold moves towards the right.
→ more observations will be classified as class w_1.

final answer: decision threshold = 4.4055, $x = 4.5$

belong to class w_2, and overall probability of error is approximately 15.38 %.

Task 9. Minimum Risk decision using Loss Matrix

Given $p(w_1|x) = 0.3$, $p(w_2|x) = 0.7$

Loss matrix:

Action	Disease (w_1)	Healthy (w_2)
Declare disease (α_1)	0	8
Declare healthy (α_2)	50	0

Step 1: calculate Conditional risk for α_1

formula: $R(\alpha_1|x) = \sum_i \lambda(\alpha_1|w_i)\, p(w_i|x)$

sub, $= 0\,(0.3) + 8\,(0.7)$

$R(\alpha_1|x) = 5.6$

Step 2: calculate Conditional risk for α_2

$R(\alpha_2|x) = 50\,(0.3) + 0\,(0.7)$

$R(\alpha_2|x) = 15$

Step 3: Select the minimum-risk action

Comparing, $R(\alpha_1|x) = 5.6$

$R(\alpha_2|x) = 15$

since, $5.6 < 15$

choose α_1 (Declare disease)

Step 4 : Compare with MAP decision

* MAP selects the class with the highest posterior probab
* Here, healthy has a probability 0.7, so MAP selects healthy.
* Minimum-risk decision selects disease because the cost of missing disease is higher.

Therefore, MAP and minimum-risk decisions are different.

Step 5 : find the posterior threshold.

Let, $P(\omega_1|x) = P$

for disease declaration to be optional,

$$8(1-P) < 50P$$

$$8 - 8P < 50P$$

$$8 < 58P$$

$$P \Rightarrow \frac{8}{58}$$

$$\boxed{P > 0.1379}$$

Correction to the threshold interpretation : the correct threshold is 0.1379 (13.79%), not 0.2759.

final Answer:

* Risk of declaring disease = 5.6
* Risk of declaring healthy = 15.
* Minimum-risk action = Declare disease
* MAP decision = Declare healthy.
* Disease declaration is optimal when posterior probability exceeds 0.1379.

K 10. Bias-Variance Trade-off and model selection

Given:

polynomial Degree	1	2	3	5	9
Training MSE	9.0	4.1	2.2	0.9	0.1
validation MSE	9.5	4.6	2.9	3.6	11.8

Step 1: calculate generalisation gap.

formula: Generalisation Gap = validation MSE - Training MSE

calculations:

for degree 1: 9.5 - 9.0 = 0.5

for degree 2: 4.6 - 4.1 = 0.5

for degree 3: 2.9 - 2.2 = 0.7

for degree 5: 3.6 - 0.9 = 2.7

for degree 9: 11.8 - 0.1 = 11.7

Generalisation gap table:

Degree	Training MSE	validation MSE	Generalisation Gap
1	9.0	9.5	0.5
2	4.1	4.6	0.5
3	2.2	2.9	0.7
5	0.9	3.6	2.7
9	0.1	11.8	11.7

Step 2: Diagnose under-fitting and over-fitting.

- Degree 1: High training and validation errors indicate under-fitting.
- Degree 2: Good performance with a small generalisation gap.
- Degree 3: Low validation error and good generalisation.
- Degree 5: Training is low, but validation error increases,

indicating over-fitting.

• Degree 9: very low training error but very high validation error, indicating severe over-fitting.

Step 3: Selects the best model

The best model is selected based on the lowest validation MSE.

Degree	validation MSE
1	9.5
2	4.6
3	2.9
5	3.6
9	11.8

Best model = Degree 3

Justification:

• Degree 3 has the lowest validation MSE is 2.9.

• It provides a good balance between under-fitting and over-fitting.

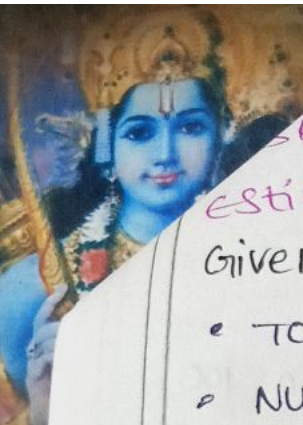
• It generalises better to unseen data.

Step 4: Recommend two techniques to improves the degree-9 model.

1. Regularisation: Apply L1 or L2 regularisation to penalise large model coefficients and reduce over-fitting.

2. Cross-validation: use K-fold cross-validation to select an appropriate polynomial degree and improve generalisation.

Final Answer: Degree 3 is the best model because it has the lowest validation MSE of 2.9. Regularisation and cross-validation can help improve the degree-9 model.



E 11. Maximum Likelihood estimation and Bayessian estimation

Given:

- Total coin tosses = 10
- Numbers of heads = 7
- Number of trails = 3
- prior distribution = Beta (2,2)

Step 1: Calculate maximum likelihood estimate (MLE).

formula $\hat{\theta}_{ML} = \frac{\text{Numbers of heads}}{\text{Total tosses}}$

$= \frac{7}{10}$

$= 0.7$

Step 2:- Calculate MAP estimate

the prior distribution is: $\theta \sim Beta(\alpha, \beta)$

$\alpha = 2, \quad \beta = 2$

The posterior distribution is:

$Beta(\alpha + h, \beta + t)$

where h = heads, t = tails

$Beta(2+7, 2+3)$

posterior = $Beta(9,5)$

MAP formula: $\hat{\theta}_{map} = \frac{\alpha' - 1}{\alpha' + \beta' - 2}$

$= \frac{9-1}{9+5-2}$

$= \frac{8}{12}$

$\hat{\theta}_{MAP} = 0.6667$

Step 3: calculate posterior mean

formula $E[\theta|D] = \frac{\alpha'}{\alpha' + \beta'} = \frac{9}{9+5} = \frac{9}{14}$

posterior mean = 0.6429

Step 4: evaluate the estimates for 70 heads in 100 tosses.

Now, $h = 70, t = 30, n = 100$

MLE: $\hat{\theta}_{ML} = \frac{70}{100} = 0.7$

posterior distribution:

Beta(2+70, 2+30)

Beta(72, 32)

MAP: $\hat{\theta}_{MAP} = \frac{72-1}{72+32-2} = \frac{71}{102}$

$= 0.6961$

posterior mean: $E[\theta|D] = \frac{72}{72+32} = \frac{72}{104}$

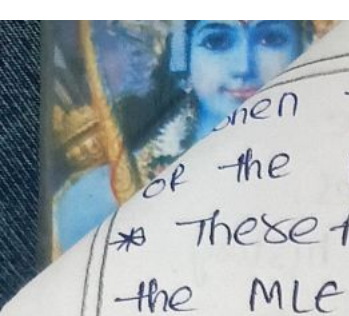
$= 0.6923$

Comparision of all estimates

estimate	7 Heads / 10 Tosses	70 Heads / 100 Tosses
MLE	0.700	0.7000
MAP	0.6667	0.6961
posterior mean	0.6429	0.6923

Step 5: Explain the role of the prior.

* The Beta (2,2) prior represents initial knowledge about the coin's probability of showing heads.
* The prior influences the MAP estimate and posterior mean.



when the number of observations increases, the influence of the prior decreases.

* Therefore, the Bayessian estimates move closer to the MLE

final Answers: for 10 tosses, MLE = 0.7, MAP = 0.6667 and Posterior mean = 0.6429. for 100 tosses, the estimate becomes 0.7, 0.6961 and 0.6923 respectively.

TASK 12. ICU sepsis easly-warning system

(a) Design the Learning system

1. TASK (T)

- predict whether an ICU patient will develop sepsis within the next 6 hours.
- It is a binary classification problem.

2. performance Measure (P):

- sensitivity to identify sepsis patients
- specificity to identify healthy patients.
- precision, recall and f1-score.
- Reduce false negatives because missing sepsis cases is costly.

3. Experience (E)

- Historical electronic health records.
- patient vital signs, laboratory values and previous medical outcomes.
- Records of patients with and without sepsis.

4. Target functions:

$$f(x) \rightarrow Y$$

where:

→ X = patients health information → Y = sepsis or no sepsis.

5. Input Representation.

* Heart rate
* Body temperature
* Blood pressure
* Respiratory rate
* Laboratory values.
* patient medical history.

6. Learning Algorithm:

→ Supervised Learning

→ Random Forest or Logistic Regression can be used for binary classification.

7. validation strategy:

- Divide the dataset into training and validation sets.
- Train the model using historical patient data.
- Evaluate using sensitivity, specificity and precision.
- use patient-level and time-based splitting to avoid data leakage.

(b) Calculate TP, FN, Fp and TN

Given

- Total patients = 100
- Sepsis prevalence = 5%
- sensitivity = 0.90
- Specificity = 0.80

step 1: calculate actual positive patients.

$$= 1000 \times \frac{5}{100} = \boxed{50}$$

step 2: calculate actual negative patients.

$$= 1000 - 50 = \boxed{950}$$

step 3: calculate True positive (TP),

$$TP = 50 \times 0.90$$

$$\boxed{TP = 45}$$

step 4 : calculate False Negative (FN)

$$FN = 50 - 45 = \boxed{5}$$

step 5 : calculate True Negative (TN)

$$TN = 950 \times 0.80 = \boxed{760}$$

step 6 : calculate False positive (FP),

$$FP = 950 - 760 = 190$$

$$\boxed{FP = 190}$$

Confusion matrix

Actual/ predicted	sepsis	NO sepsis
sepsis	TP=45	FN=5
NO sepsis	Fp=190	TN=760

(c) calculate PPV and NPV

1. positive predictive value(PPV):

formula:

$$PPV = \frac{TP}{TP + FP} = \frac{45}{45+190} = \frac{45}{235}$$

$$\boxed{PPV = 19.15\%}$$

2. Negative predictive value (NPV):

formula:

$$NPV = \frac{TN}{TN+FN} = \frac{760}{760+5} = \frac{760}{765}$$

$$\boxed{NPV = 99.35\%}$$

(d) Compare the total cost of three strategies.

Given:

- cost of missed sepsis case = ৳100.
- cost of false alarm = ৳5.

1. Using the trained model:

missed cases = 5

false alarms = 190

Cost = (FN × 100) + (FP × 5)

= (5 × 100) + (190 × 5) = 500 + 950

Cost = ৳ 1450

2. Never alerting

- All 50 sepsis patients are missed
- No false alarms occur.

Cost = 50 × 100

Cost = ৳ 5000

Cost = ৳ 5000

3. Never alerting every patient

- All 50 sepsis patients are detected.
- 950 healthy patients receive false alerts.

Cost = 50 × 950

Cost = ৳ 4750

Comparision Table

Strategy	Total Cost
Using trained model	৳ 1,450
Alerting every patient	৳ 4,750
Never alerting	৳ 5,000

Analysis: Using the trained model has the lowest total cost compared with the other two strategies.

Recommend a Deployment policy

- Deploy the model as a clinical decision-support system.
- Generate alerts when the predicted risk of sepsis crosses the selected threshold.
- Since missing a sepsis case costs ₹100 and a false alarm costs only ₹5, the system should prioritise detecting sepsis cases.
- The alert threshold should be selected using Bayes decision theory.

Bayes decision rule:

Let p be the posterior probability of sepsis.

$$\text{Cost (Alert)} = 5(1-p)$$

$$\text{Cost (No Alert)} = 100p$$

for alerting to be optimal,

$$5(1-p) < 100p$$

$$5 - 5p < 100p$$

$$5 < 105p$$

$$\boxed{p > 0.0476}$$

Therefore, the theoretical minimum-cost decision threshold is approximately 4.76%.

Final answer: The hospital should deploy the model with a cost-sensitive alert threshold, prioritising early detection of sepsis. Clinical validation and doctor supervision are also necessary before deployement.